

ANNUAL ACTIVITY REPORT (FINANCIAL YEAR 2025)

AETERNA CORPORATE & OPERATIONAL REPORT

1. Executive Summary

AETERNA is an advanced R&D; technology firm specializing in zero-entropy software architectures, kernel-level eBPF security frameworks, and high-performance Edge AI systems for critical national infrastructure. Throughout 2025, AETERNA solidified its role as a key contributor to European digital sovereignty, delivering high-speed telemetry platforms and securing strategic industrial control assets.

2. Key Scientific & R&D; Accomplishments

- **Mojo Vectorized Classifier:** Completed the development of the $O(1)$ vectorized neural signal separation core, achieving 35,000x execution speed-ups compared to legacy python DSP structures.
- **eBPF Sentinel Shield:** Designed and deployed clean-room Linux kernel security containment loops, proving lateral SCADA isolation speeds under 1.02ms.
- **SOP Stream Acquisition Core:** Built zero-latency light polarization shift data acquisition pathways, successfully tested on subsea active fiber trunks.

3. Active R&D; Projects Portfolio

During 2025, AETERNA actively coordinated and executed 3 major European-funded and national research initiatives:

- *EIC Accelerator - AETERNA Core Node* (Grant Agreement 865986222) - €4.2M total budget.
- *Horizon Europe Critical Infrastructure - CIRCAT-VIVI* - €2.8M total budget.
- *Horizon Europe Smart Cities - AETERNA-AIGIS* - €5.1M total budget.

4. Corporate Governance & Financial Standing

AETERNA operates under strict corporate compliance and zero foreign debt. The company's assets grew by 24% in 2025, driven by sovereign software licensing and dedicated research grants. AETERNA's operational facilities in Pomorie, Bulgaria, are certified under ISO/IEC 27001 for information security management, meeting all critical security toolbox criteria established by the EU.

Certified by:

Name: Dimitar Prodrumov

Title: Sovereign Systems Architect & Legal Rep

Secure Operations Signature: 0xCF_AAR_AETERNA_2025_APPROVED